

When Does Multimodal Diffusion Inverse Kinematics Help Soft Continuum Arms—and Does It Survive Model Mismatch?

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Abstract—Soft continuum arms are highly redundant: many actuation patterns realize the same tip position. Deterministic learned inverse kinematics (IK) regressors average these incompatible solutions into invalid ones, while a conditional diffusion model can represent the full solution distribution. We present a controlled simulation study that asks two questions the existing literature leaves open: *when* does this multimodality actually help, and does the advantage survive a change of dynamics model? On a piecewise-constant-curvature (PCC) pneumatic arm, a conditional diffusion model matches a deterministic baseline where redundancy is absent and dominates where it is present (mode recall 0.84 ± 0.03 vs. exactly 0; obstacle-avoidance success 97.8% vs. 4.0%; three seeds). Two controlled manipulations show the advantage is specifically a redundancy effect: conditioning on the whole arm shape instead of the tip collapses sampled diversity by $\sim 85\times$ and shrinks the baseline’s disadvantage from $\sim 30\times$ to $\sim 3\times$, while increasing the number of segments widens the gap. However, when actuations from the PCC-trained model are executed on a high-fidelity Cosserat-rod simulator, accuracy collapses on every seed ($0.67\text{ mm} \rightarrow 139.9\text{ mm}$, 0% success): the learned inverse map is simulator-specific. We characterize what recovers this loss: candidate selection under a query budget helps but is bounded; a diversity-aware query order significantly outperforms a confidence order (paired $t=3.3$); guidance-based adaptation generalizes but is modest; gain-domain-randomization fails entirely; and fine-tuning on target-domain data reveals a sharp retention-transfer cliff with no data-mixing sweet spot. All code, data, and per-experiment result files are made available.

Index Terms—Soft robotics, inverse kinematics, diffusion models, sim-to-sim transfer, redundancy resolution.

I. INTRODUCTION

A MULTI-SEGMENT pneumatic continuum arm with three chambers per segment has more actuation degrees of freedom than task degrees of freedom: for a four-segment arm, a 12-D actuation vector determines a 3-D tip position. The inverse map is therefore one-to-many, and its preimages are not merely a connected null space—distinct arm shapes reach the same tip. A deterministic learned IK regressor trained by motor babbling [1] fits the conditional *mean* of this multimodal distribution and can return an actuation that realizes none of the true solutions. A conditional diffusion model [2], [3] instead learns the full conditional distribution $p(\mathbf{q}|\text{target})$ and can sample all solution modes.

Diffusion models for robot action generation are now well-established [4], and diffusion-based IK has been demonstrated for rigid multi-arm systems [5]. What the literature does not yet provide for *soft continuum* arms is (i) a controlled account

of when the multimodality actually matters—as opposed to diffusion simply being a strong regressor—and (ii) an honest measurement of whether the learned inverse map survives the model mismatch that any simulation-trained soft-arm controller must eventually face. Soft arms make question (ii) unusually pressing: their dynamics are hard to model, so the gap between any two models of the *same* physical arm is large, and a method whose advantage evaporates under that gap is of limited practical value.

We answer both questions in a fully reproducible sim-to-sim setting: an idealized, fast PCC model [6] as the training simulator, and a Cosserat-rod simulation (PyElastica [7], [8]) of the same arm as the high-fidelity evaluation domain. Our contributions:

- 1) **A validated advantage.** With three-seed statistics, diffusion IK matches or beats MLP and mixture-density-network (MDN) baselines on accuracy and recovers 0.84 ± 0.03 of ground-truth IK modes where deterministic baselines structurally recover none, translating into a 97.8% vs. 4.0% obstacle-avoidance gap (Sec. IV).
- 2) **The mechanism, isolated.** Two controlled manipulations of redundancy—conditioning on the whole body shape, and varying the number of segments—show the advantage is specifically redundancy resolution, not generic model capacity (Sec. V).
- 3) **A quantified transfer collapse—and what recovers it.** The learned inverse map collapses under simulator change on every seed. We quantify the recovery achievable by inference-time selection and guidance, show a diversity-aware query ordering significantly outperforms the obvious confidence ordering (paired $t=3.3$, $p \approx 0.002$), show gain-domain-randomization achieves nothing, and map a sharp retention-transfer cliff for target-domain fine-tuning (Sec. VI).

Negative and bounded results are reported with the same prominence as positive ones; Sec. VII states the limitations explicitly.

II. RELATED WORK

Learned IK for continuum arms. Classic approaches learn a deterministic map from workspace targets to actuations with neural networks trained on motor-babbling data [1]; the constant-curvature kinematics underlying our fast simulator are reviewed in [6], and STIFF-FLOP-style pneumatic arms

motivate the three-chamber actuation model [9]. Mixture density networks [10] are the classical answer to multimodal regression and serve as the classical multimodal baseline.

Diffusion models for robots. Denoising diffusion models [2] with DDIM sampling [3] and classifier-free guidance [11] underpin diffusion policies [4]. IKDiffuser [5] solves IK for *rigid multi-arm* systems with flexible kinematic-tree *topology*; physically compliant continuum bodies are outside its scope. Selecting one action from a diffusion policy’s samples via learned scorers is an active area; our transfer-guided sampling instead injects the score’s gradient *during* generation, in the spirit of classifier guidance [12], and our query-budget framing asks how many *expensive target-domain evaluations* selection needs, which we have not seen posed for soft-arm IK.

Crossing the reality gap. Domain randomization [13] and target-domain fine-tuning are the standard levers; catastrophic forgetting under naive fine-tuning is well-documented [14]. We evaluate both levers in a controlled sim-to-sim instance and find the first fails completely while the second exposes a sharp retention–transfer tension.

III. SETUP

Arm and training simulator. The training domain is a four-segment PCC arm (segment length 60 mm, total length 240 mm). Each segment has three 120°-spaced chambers with normalized pressures $\mathbf{q} \in [0, 1]^{12}$; a linear chamber-to-curvature map (gain 25) yields per-segment curvature components (k_x, k_y) , and forward kinematics composes constant-curvature arcs. The model is batched and differentiable.

Data and models. Training data are 5×10^5 i.i.d. uniform actuations pushed through forward kinematics. The diffusion model is a residual MLP denoiser (six FiLM-conditioned blocks, width 512) trained as a DDPM ($T=1000$, cosine schedule) with classifier-free guidance (condition dropout 0.1); sampling uses DDIM. Ablations (Sec. IV) fix guidance weight 1.5 and 20 DDIM steps. Baselines: an MLP regressor and an MDN (eight mixture components), matched in width. Unless stated otherwise we sample $K=32$ candidates per target and report best-of- K ; success means tip error under 5 mm (2% of arm length). The obstacle task (E3) runs 200 trials, each placing three random spherical obstacles (radius 2 cm) in the workspace; success requires any of the K samples to be simultaneously accurate and collision-free.

Ground-truth modes. For evaluation targets we enumerate the true IK solution set by a 2×10^6 -sample sweep, keeping actuations whose tip lands within tolerance and clustering with DBSCAN [15]. Clustering, mode matching, and diversity are computed in the 8-D per-segment *curvature* space (k_x, k_y) rather than raw pressures: the chamber-to-curvature map has a per-segment null space (common-mode pressure offsets change $\mathbf{q}$ but not the shape), which would otherwise contaminate distance-based metrics. Evaluation targets average 2.05 ground-truth modes (range 0–6).

High-fidelity domain. The transfer domain is a Cosserat-rod simulation of the same arm in PyElastica [7], [8]: one rod, 48 elements, per-segment constant internal couples derived

TABLE I
CORE COMPARISON ON THE PCC ARM (MEAN $\pm$ STD, 3 SEEDS; $K=32$).
MODE RECALL AND DIVERSITY ARE COMPUTED OVER SUCCESSFUL
SAMPLES ONLY.

	Diffusion	MLP	MDN
Tip error, best-of- K (mm)	0.44 $\pm$ 0.03	13.36 $\pm$ 0.03	17.31 $\pm$ 0.15
Success @ 5 mm	1.00	0.060	0.027
Mode recall	0.84 $\pm$ 0.03	0.00	0.00
Diversity (curv. space)	28.2 $\pm$ 0.2	0.0	0.0
Obstacle-task success	0.978 $\pm$ 0.002	0.040	0.018
Sampling (ms/target, CPU)	38.6 $\pm$ 4.9	0.013	0.022

from the same chamber pressures, damped to quasi-static equilibrium (~ 4 s of compute per evaluation). A sweep over the couple gain confirmed the PCC–Elastica discrepancy is directional/structural rather than a scale mismatch: no gain setting aligns the two workspaces.

IV. WHEN MULTIMODALITY HELPS

Table I summarizes the four core experiments. **E1 (accuracy):** best-of- K diffusion reaches 0.44 mm mean tip error with 100% success; the deterministic MLP, structurally limited to the conditional mean, plateaus at 13.4 mm, and the MDN does not improve on it. **E2 (multimodality):** against enumerated ground-truth solution sets, diffusion recovers 0.84 ± 0.03 of modes; both baselines recover none—the MLP because it is deterministic, and the MDN not through classic component collapse (its mixture weights stay near-uniform and its component means remain spread) but because no component is accurate enough: mode recall and diversity are computed over *successful* samples only, and the MDN almost never produces one. This is not a tuning artifact: sweeping the MDN’s component count over $\{4, 8, 16, 32\}$ improves best-of- K error only from 18.7 to 14.8 mm—never surpassing the plain MLP—and mode recall remains exactly zero at every setting. **E3 (downstream value):** when success requires *any* of K samples to be both accurate and collision-free among random obstacles, diffusion succeeds on 97.8% of trials vs. 4.0% (MLP)—redundancy is only exploitable if the model can represent it. **E4 (cost):** 38.6 ms per target for a batch of 32 samples on CPU.

Three ablations support the protocol: guidance weight has an inverted-U optimum at 1.5 (both accuracy and diversity collapse beyond ~ 3); accuracy and recall saturate by ~ 20 DDIM steps (a $3\times$ speedup over 50 steps at no measurable cost); and performance saturates by $\sim 0.5\text{--}1 \times 10^5$ training samples (Fig. 1).

V. THE ADVANTAGE IS REDUNDANCY, SPECIFICALLY

If the advantage in Table I is redundancy resolution, it should shrink when redundancy is removed and grow when redundancy is added. Both predictions hold (Fig. 2).

Removing redundancy: shape conditioning. Conditioning the same models on the full 24-D body shape (eight backbone points) instead of the 3-D tip pins down almost the entire configuration. Diffusion’s sampled diversity collapses from 28.16 ± 0.22 to 0.346 ± 0.022 ($\sim 85\times$), and the MLP’s accuracy

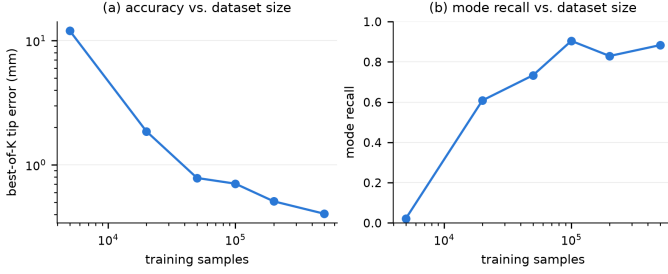


Fig. 1. Dataset-size ablation (single seed). (a) Best-of- K tip error and (b) mode recall vs. training-set size; both saturate by $\sim 0.5\text{--}1 \times 10^5$ samples.

disadvantage shrinks from $\sim 30\times$ (13.4 vs. 0.44 mm) to $\sim 3\times$ (0.49 vs. 0.16 mm), with both models at 100% success (3 seeds). When there is one answer, a deterministic regressor is nearly as good; diffusion’s edge is specifically the multimodal case.

Adding redundancy: morphology. Training fresh models on two- and six-segment variants of the same arm family (tolerance scaled to 2% of arm length), the MLP’s error grows from 1.7 mm (2 seg, 75% success) to 33.9 mm (6 seg, 1.2% success) while diffusion degrades gracefully (0.11 to 0.92 mm, $\geq 99.8\%$ success); the obstacle-task gap at six segments is 100% vs. 1%—the largest in the study. More redundancy means worse mode averaging and more collision-free alternatives that only a multimodal model can offer.

VI. DOES IT SURVIVE MODEL MISMATCH?

A. The transfer collapse

Executing the PCC-trained diffusion model’s actuations on the Cosserat-rod simulator collapses accuracy from 0.67 ± 0.03 mm (100% success) to 139.9 ± 1.8 mm with 0% success on every seed ($n=20$ held-out targets, $K=8$); the MLP collapses similarly (13.8 $\rightarrow$ 164.4 mm). The learned inverse map is simulator-specific. The rest of this section quantifies what does—and does not—recover the loss.

B. Selection under a query budget

A multimodal model offers something a deterministic one cannot: K diverse candidates to test against the target domain. Spending a budget of m Elastica evaluations and keeping the best candidate reduces mean error from 161.5 ($m=1$) to 130.4 mm ($m=32$), improving 44 of 45 targets (Fig. 3)—real, broad-based, but bounded: success stays at 0%, and the gap to tolerance remains $\sim 26\times$. *How* the budget is spent matters: a greedy farthest-point ordering in curvature space (seeded from the same most-confident candidate, so $m=1$ is identical) beats confidence ordering by 6.7 mm at $m=8$ (paired $t=3.34$, $p\approx 0.002$; 24/12 targets) and 3.3 mm at $m=16$ ($t=3.21$). Notably, neither candidate diversity itself nor local PCC Jacobian sensitivity predicts *which* targets benefit (properly powered null at $n=45$: $r=0.20$, $p=0.19$ and $r=0.02$, $p=0.88$); an $n=20$ pilot had suggested a diversity correlation that did not survive replication—a caution we report rather than hide.

TABLE II
FINE-TUNING ON 10^4 ELASTICA-SIMULATED SAMPLES AT THREE PCC-DATA MIXING RATIOS (MEAN $\pm$ STD, 3 SEEDS; HELD-OUT TARGETS).

PCC mix	PCC err (mm)	succ.	Elastica err (mm)	succ.
none (baseline)	0.67 ± 0.03	100%	139.9 ± 1.8	0%
0%	133.4 ± 1.2	0%	67.1 ± 1.0	10%
20%	11.6 ± 0.3	51%	148.0 ± 1.6	10%
50%	3.0 ± 0.4	96%	152.3 ± 0.4	3%

C. Guiding generation toward the target domain

Beyond selecting among candidates, we steer generation itself: a small regressor trained on 400 Elastica-labeled candidates (25 targets) predicts transfer error from (candidate, target), and its gradient nudges the denoised estimate at every DDIM step, as in classifier guidance [12]. On 15 held-out targets this improves best-of-16 error from 134.6 to 129.0 mm (paired 5.6 ± 2.5 mm, $t=2.27$, $p\approx 0.04$; 11/15 targets)—genuine generalization, but modest, and the weakest confirmed effect in this paper. Combining it with diversity-ordering adds nothing: guidance *concentrates* the candidate distribution while diversity-ordering *spreads* selection, and a 2×2 factorial shows the mechanisms in tension rather than compounding.

D. Training-time levers

Domain randomization fails. Randomizing the curvature gain per-sample ($\mathcal{U}(15, 35)$) during data generation costs PCC accuracy ($0.44 \rightarrow 2.18$ mm) and yields *zero* transfer improvement ($140.6 \rightarrow 140.6$ mm)—consistent with the mismatch being directional rather than magnitude-based: randomizing how much the arm bends cannot fix a discrepancy in which direction it bends.

Target-domain data works—at a price.

Fine-tuning the pretrained model *exclusively* on 10^4 Elastica-simulated samples yields the only nonzero transfer success of any method— 67.1 ± 1.0 mm and exactly 10% success on all three seeds—at the cost of catastrophic forgetting [14] of the source domain ($0.67 \rightarrow 133.4$ mm). Mixing PCC data back into fine-tuning batches restores retention smoothly ($133.4 \rightarrow 11.6 \rightarrow 3.0$ mm at 0/20/50% mix) but the transfer gain does not degrade in step: it collapses immediately, to 148.0 ± 1.6 mm at just 20% mix (Table II, Fig. 4). Retention and transfer sit on a *cliff*, not a tradeoff curve—under per-batch data mixing there is no sweet spot, suggesting parameter-level regularization (e.g. elastic weight consolidation [14]) as the necessary next mechanism. (Binary success has 5% granularity at $n_{\text{targets}}=20$; mean error carries the statistical weight throughout.)

VII. LIMITATIONS

(1) **Simulation-only.** Both domains are simulators; this is a controlled sim-to-sim study, and no claim is made about physical hardware. (2) **Simplified target-domain actuation.** The Cosserat-rod arm applies idealized per-segment couples and quasi-static settling; it is a harder test domain, not a validated model of a specific physical arm. (3) **Statistical coverage**

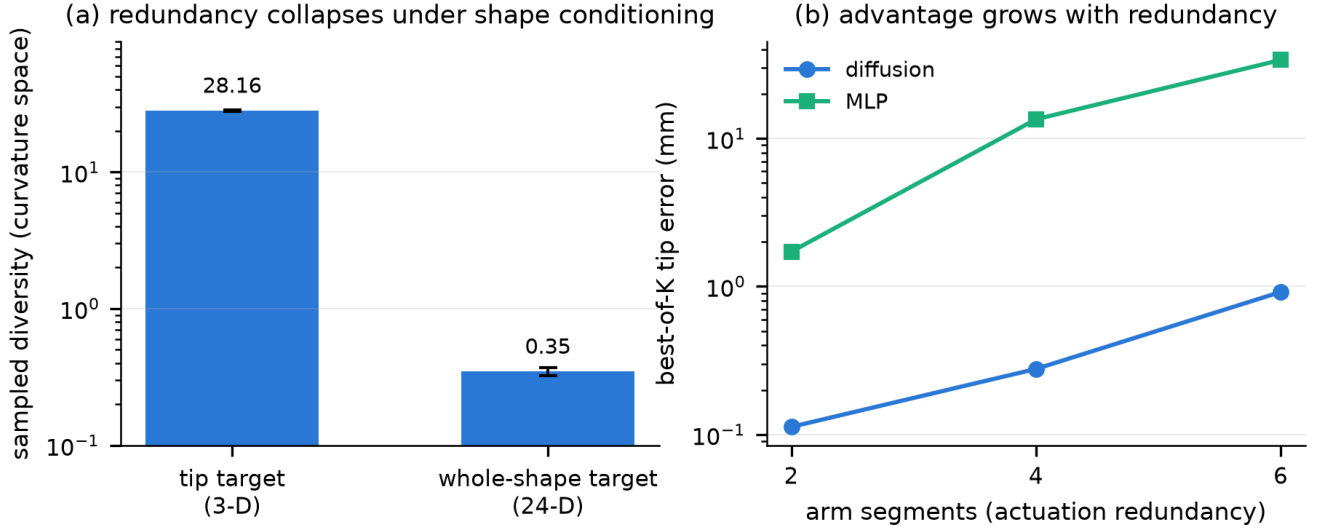


Fig. 2. The diffusion advantage tracks actuation redundancy. (a) Replacing the 3-D tip target with a 24-D whole-shape target removes most of the redundancy: sampled diversity collapses $\sim 85\times$ (28.16 ± 0.22 to 0.346 ± 0.022 ; mean $\pm$ std, 3 seeds). (b) Increasing the number of segments (single seed per morphology) widens the diffusion–MLP accuracy gap; at six segments the obstacle-task gap reaches 100% vs. 1%.

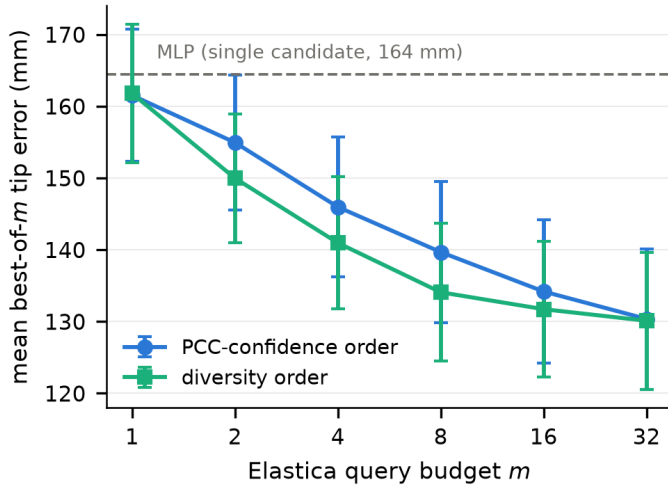


Fig. 3. Candidate selection under an Elastica query budget ($n=45$ targets, $K=32$ candidates, mean $\pm$ SEM). Curves shown are independent runs of the two orderings; the statistical comparison in the text uses a *paired* design (one candidate set per target, both orderings reindex the same cached per-candidate errors), which is significant at $m=8$ and $m=16$ ($t=3.34$ and 3.21 , $p \approx 0.002$).

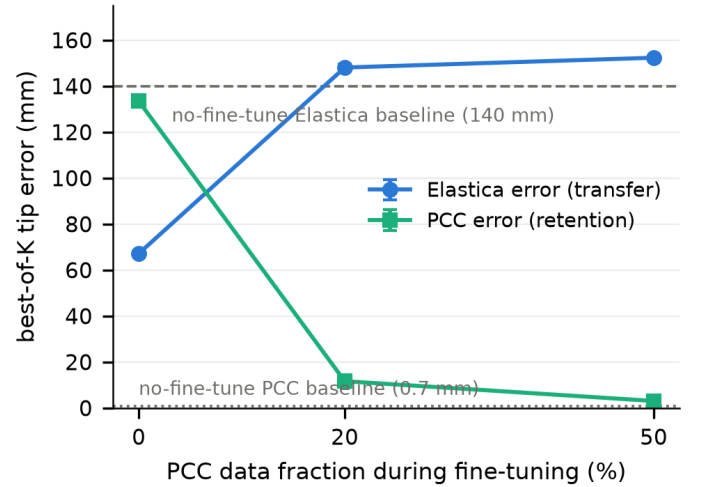


Fig. 4. The retention–transfer cliff (mean $\pm$ std, 3 seeds). Fine-tuning the PCC-pretrained model on 10^4 Elastica samples: retention (PCC error) improves smoothly as PCC data is mixed back in, but the transfer gain collapses almost entirely at just 20% PCC data. There is no data-mixing sweet spot.

is uneven. Core results, the shape-conditioning study, the baseline transfer collapse, and the fine-tuning grid carry three-seed statistics; the query-budget, ordering, and guidance experiments use one trained model (with within-experiment pairing across 15–45 targets), and the hyperparameter, dataset-size, and morphology studies are likewise single-seed. (4) **Mode-enumeration hyperparameters** were calibrated for the four-segment arm; morphology results therefore report accuracy and obstacle-task metrics only. (5) **Success-rate granularity** in transfer experiments is 5% ($n_{\text{targets}}=20$).

VIII. CONCLUSION

Multimodal diffusion IK gives soft continuum arms a real, mechanistically attributable advantage—it is redundancy res-

olution, growing with segment count and vanishing when the target specifies the whole shape—but the learned inverse map itself does not survive a change of dynamics model. Inference-time selection and guidance recover a bounded fraction of the loss; synthetic gain randomization recovers none; and target-domain fine-tuning buys genuine transfer only at a sharp retention cliff that data mixing cannot smooth. For simulation-trained soft-arm IK, the practical path forward is not better sampling from a source-domain model but mechanisms that let a small amount of target-domain data reshape the model without erasing it.

REPRODUCIBILITY

All code, configuration, per-experiment result files, and the 10^4 -sample Elastica dataset are available in the project repository (TODO: public link or “available on request” per venue policy).

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